Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/26/2009
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Project Title:	Simulation of Combustion Systems with Realistic G-Jitter		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	84112 - 9003	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	07/01/2004	**End Date:**	09/30/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
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	In the proposed study, which is a renewal of an existing project, a transient, fully three-dimensional simulation code will be developed to simulate the effects of realistic g–jitter on a number of combustion systems. Acceleration disturbances of realistic orientation, magnitude and time dependence can be easily included in the simulation. A number of microgravity combustion experimentalists have agreed to use a preliminary version of the code. The experience of these users will be essential to both fully debugging the code and making it more user friendly. At the end of the project the code will be made available for general use by the microgravity combustion community. The simulation algorithm is based on techniques used in an existing large eddy simulation code which has successfully simulated fire dynamics in complex domains. This technique is very efficient computationally so that simulations of the entire experimental domain with one to two million grid cells can be performed on current single processor workstations. An efficient cell blocking method allows the user to easily define and modify the solid surfaces		

Task Description:	comprising the simulation domain and diagnostic equipment. To preserve computational efficiency, simple combustion models are used (e.g., flame sheet or highly reduced, finite rate, second order, global Arrhenius kinetics). A multiblocked grid approach will be implemented to allow regions requiring high resolution (i.e., the flame) to be covered by a fine grid while coarser grids cover the rest of the experimental volume. Acceleration disturbances or g–jitter about the background level of reduced gravity exist in all the microgravity facilities. Low frequency g–jitter has been repeatedly observed to disturb a number of combustion systems. Guidelines regarding tolerable levels of acceleration disturbances for a given combustion system (i.e., flame balls, gas jet diffusion flames, etc.) have been developed for use in the design of ISS experiments. The validity of these guidelines, however, remains unknown. The proposed simulation code will allow users to assess the g-jitter effects on their flame for the microgravity facility in question. This will help investigators plan experiments and interpret their data. For example on the ISS an acceptable acceleration environment will be achieved through the combined use of a passive isolation system and scheduling appropriate crew/station activity (e.g., no ergometer exercise). Different flames are expected to have different acceleration environment requirements and, therefore, demands on crew/station activity. The proposed simulation code will provide one means of determining these requirements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2004.
Bibliography Type:	Description: (Last Updated:)